BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 2 | Semester: Spring 2025
Duration: 25 minutes | Full Marks: 15

CSE 340: Computer Architecture

Name: Solution	ID:	Section: 2

1. if (a > b and b < c): ($c1$, $c2$)
  a = b
elif (a == 5):
  c = 10
else:
  d = d + 10

Construct the equivalent RISC-V code of the above high-level code. Variables a,b,c,d are associated with register X_{11}, X_{12}, X_{13}, X_{14} respectively [4]

Answer:

```
BGE X12, X11, Elif
BGE X12, X13, Elif
ADD X11, X12, X0
Beq X0, X0, Exit

Elif:
Add X5, X0, 5
BNE X11, X5, Else
Addi X13, X0, 10
Beq X0, X0, Exit

Else:
Addi X14, X14, 10
Exit:
```

2. In the RISC-V architecture, each instruction is encoded as a 32-bit binary sequence. In this standard format, the rs (source register) and rd (destination register) fields are typically 5 bits in size.

Suppose we are working with a prototype version of RISC-V that includes 60 registers. For this prototype version, what would be the required size of the rs or rd field? [3]

Answer: 60 registers. register number → (0-59)
We store register number in rs/rd field. Max number is 59 → $\underbrace{111011}$ → 6 bits to represent 59.
So, size of rs/rd field = 6 bits.

3. Find the RISC-V code of the following machine Codes.

i. 0x00AA9B93 [3]

Answer: Slli X23, X21, 10

$\underbrace{0000\ 000}_{f6}\underbrace{0\ 1010}_{imm}\ \underbrace{1010\ 1}_{rs\text{-}1}\underbrace{001}_{f3}\ \underbrace{1011\ 1}_{rd}\underbrace{001\ 0011}_{opcode}$ → I type

f3 → to find exact ins.

ii. 0x08C2BAB3 [3]

Answer:

$\underbrace{0000\ 100}_{f7}\underbrace{0\ 1100}_{rs2}\ \underbrace{0010\ 1}_{rs1}\underbrace{011}_{f3}\ \underbrace{1010\ 1}_{rd}\underbrace{011\ 0011}_{opcode}$ → R type

OR X21, X5, X12

combination of f3 & f7 to find exact instruction.

4. Although SRLI is an I-type instruction, the immediate field is restricted to 6 bits. Explain the reason for this constraint? [2]

Answer: Register size = 64 bits. So, valid shifting 1 to 63

SRLI X_{12}, X_{13}, ___

How many times we shift.

Maximum shifting 63 times.

$63 = \underbrace{111111}_{6 \text{ bits}}$

Format	Instruction	Opcode	Funct3	Funct7/Funct6
R	ADD	0110011	001	0000000
	SUB	0110011	110	0000001
	AND	0110011	010	0000001
	OR	0110011	011	0000100
	SLL	0110011	101	0000100
I	LD	0000011	000	N/A
	LW	0000011	010	N/A
	LH	0000011	011	N/A
	LB	0000011	001	N/A
	ADDI	0010011	000	N/A
	SLLI	0010011	001	000000
	SRLI	0010011	010	000000
	ORI	0010011	011	N/A
S	SD	0100011	000	N/A
	SW	0100011	001	N/A
	SH	0100011	010	N/A
	SB	0100011	011	N/A